Walid Hemati

Senior Developer & Data Analyst · Shikenso

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PROFILE

Software engineer for video data pipelines and computer vision. At Shikenso I brought YOLOv8 into the detection pipeline, built a live-TV recording system from scratch and 30+ broadcaster integrations, and run the services behind them in production. I also build privacy-first Android apps and GPU scientific code. B.Sc. in Computer Engineering with Honors, Istanbul Kültür University.

EXPERIENCE

09/2024 – now

Senior Developer & Data Analyst, Shikenso

Sponsorship analytics: computer vision on broadcast and streamed sports video.

- Brought YOLOv8 into the production detection pipeline: wrote the tracker for box and polygon (segmentation) models, integrated and tested it. Detection got roughly 2× faster, and it is still the base of today's detector.
- Built training and evaluation data for the logo-detection models: VIA-to-YOLO polygon conversion, annotated evaluation sets, annotation QA tools.
- Designed and built a live-TV recording system from the first commit: about 50 channels in three countries, rolling 30-minute segments, multi-source failover, EPG tagging, a FastAPI scheduler and about 900 tests.
- Built 30+ capture integrations for broadcasters, streaming services and sports platforms across Europe, Australia and the Middle East.
- Created a TV-ratings platform for 10 audience-measurement sources, and rebuilt a panel-audience service to run about 22× faster.
- Run 36 scheduled capture services in production with Jenkins CI/CD, systemd, Kubernetes/Helm and Sentry, with regression tests for every incident fix.

07/2021 – 09/2021

Web developer intern, Miss Diamond, Istanbul

Developed the company website; maintained its computers and website.

EDUCATION

07/2019 – 07/2023

Istanbul Kültür University, Istanbul

B.Sc. Computer Engineering, with Honors

2017 – 2019

Beşir Balcıoğlu Anadolu Lisesi, Istanbul

High school

2008 – 2017

Sharjah Indian School, Sharjah, UAE

School

PROJECTS

Open source

pylbm · Scientific computing

A GPU Lattice-Boltzmann solver in PyTorch/CUDA for the D2Q9 lid-driven cavity, validated against Ghia et al. (1982).

Python · PyTorch · CUDA

In development

SteuerApp · Android · Privacy-first

An offline German tax recorder for Android. Receipts are read and classified on the device; by default nothing leaves the phone.

Kotlin · Android · On-device OCR · Room + SQLCipher

Android

Flatsnap · Android · On-device OCR

An on-device document scanner for Android. It finds the page edges, corrects the perspective and reads the text with ML Kit, all on the phone.

Android · ML Kit

Research

Emotion from EEG · Research · ML

Biosignal-based latent factor decoding for emotion detection with a variational autoencoder, on the DEAP and SEED datasets.

Python · MATLAB

Open source

Turkish review classification · NLP

Classifying Turkish Hepsiburada reviews, comparing logistic regression and SVM with BERTurk and DistilBERT.

Python · Jupyter · BERT

Coursework

yamd · Scientific computing · C++

Coursework for the university course Molecular Dynamics in C++, built on the course's Meson template with Eigen and GoogleTest.

C++ · Eigen · gtest · Meson

Review paper

Machine Learning Based Android Malware Detection: A State of Art

A review of machine-learning approaches to detecting malware on Android.

SKILLS

Languages	Python · Kotlin · TypeScript / JavaScript · C++ · SQL
Backend & data	FastAPI · Pydantic · MongoDB · PostgreSQL · pandas / NumPy
Video	ffmpeg (HLS, MPEG-TS) · yt-dlp · streamlink · Playwright · EPG data
ML & vision	PyTorch · YOLOv8 (Ultralytics) · OpenCV · ML Kit · Transformers · CUDA
DevOps	Jenkins · Docker / Podman · Kubernetes / Helm · systemd · Sentry · Git
Mobile	Android · Jetpack Compose · Room / SQLCipher · on-device ML

LANGUAGES

Dari (native) · English (C1) · Persian (B2 listening, B1 spoken) · Turkish (B1) · Urdu / Hindi (B1 spoken) · Arabic (A1–A2)